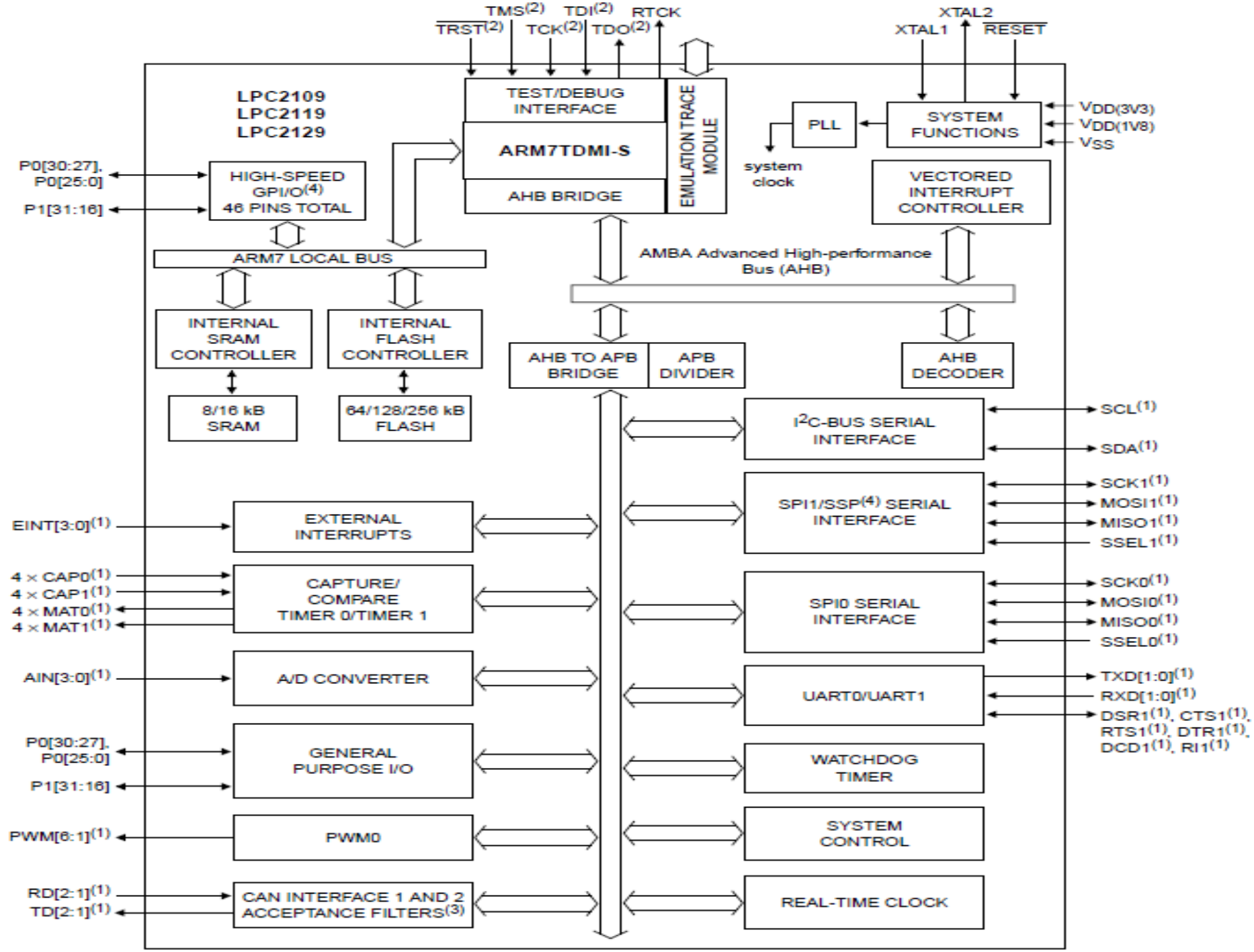

LPC2109
LPC2119
LPC2129
TMS(2)
TDI(2)
RTCK
TRST(2)
TCK(2)
TDO(2)
XTAL2
XTAL1
RESET
TEST/DEBUG INTERFACE
ARM7TDMI-S
AHB BRIDGE
EMULATION TRACE MODULE
PLL
SYSTEM FUNCTIONS
VDD(3V3)
VDD(1V8)
VSS
system clock
VECTORED INTERRUPT CONTROLLER
P0[30:27], P0[25:0]
P1[31:16]
HIGH-SPEED GPI/O(4) 46 PINS TOTAL
ARM7 LOCAL BUS
AMBA Advanced High-performance Bus (AHB)
INTERNAL SRAM CONTROLLER
INTERNAL FLASH CONTROLLER
AHB TO APB BRIDGE
APB DIVIDER
AHB DECODER
8/16 kB SRAM
64/128/256 kB FLASH
I2C-BUS SERIAL INTERFACE
SCL(1)
SDA(1)
EINT[3:0](1)
EXTERNAL INTERRUPTS
SPI1/SSP(4) SERIAL INTERFACE
SCK1(1)
MOSI1(1)
MISO1(1)
SSEL1(1)
4 × CAP0(1)
4 × CAP1(1)
4 × MAT0(1)
4 × MAT1(1)
CAPTURE/ COMPARE TIMER 0/TIMER 1
SPI0 SERIAL INTERFACE
SCK0(1)
MOSI0(1)
MISO0(1)
SSEL0(1)
AIN[3:0](1)
A/D CONVERTER
UART0/UART1
TXD[1:0](1)
RXD[1:0](1)
DSR1(1), CTS1(1), RTS1(1), DTR1(1), DCD1(1), RI1(1)
P0[30:27], P0[25:0]
P1[31:16]
GENERAL PURPOSE I/O
WATCHDOG TIMER
PWM[6:1](1)
PWM0
SYSTEM CONTROL
RD[2:1](1)
TD[2:1](1)
CAN INTERFACE 1 AND 2 ACCEPTANCE FILTERS(3)
REAL-TIME CLOCK

LPC2129 (ARM-7 Based)uC : Feature Set

1)ARM7 (TDMI-S based high-performance single core,32-bit RISC Microcontroller

2)256KB on-chip Flash ROM with ISP

3) 256K Flash ROM ,

256*1024= 262144bytes/40000h bytes(0x00000000 to 0x0003ffff)

4)16KB SRAM,

16 * 1024= 16384bytes/4000h bytes (0x40000000 to 0x40003fff)

5)General purpose I/O pins (upto 46).

6)CPU clock up to 60 MHz,

7)On-chip crystal oscillator

8)On-chip PLL

9)Two Onchip 32-bit timers (Timer0 & Timer 1) ,

10)Two UARTs(UART0 & UART1),

LPC2129 Features Cont’d

10)I2C serial interface,

11)2 SPI serial interfaces(SPI0 & SPI1)

12)Real Time Clock(RTC),

13)Watchdog Timer,

14)4-channels Onchip 10bit ADC,

15)PWM unit with up to 6 PWM outputs,

16)2 CAN channels(CAN1 & CAN2).

17)Vectored Interrupt Controller with upto 32 interrupts available,